BK.

CLINICAL REASONING

Mark’s six-week program: clinical reasoning

Written justification for the exercise prescription

Applied degree project. Simulated client with type 2 diabetes and mild right knee osteoarthritis. The accompanying program is available as a spreadsheet.

REASONING

I used Mark’s GP clearance to plan gradual exercise. He has type 2 diabetes, five years without structured training and knee pain descending stairs, but already walks his dog for 15 minutes. I built on that habit rather than assuming he could tolerate vigorous training.

My first decision was between running or sport-style intervals and lower-impact walking and cycling. I chose flat walking and a stationary bike because they allow conversational effort and can be shortened or substituted if knee symptoms increase. Aerobic work progresses from 12 to 25 minutes per gym session. I rejected hard intervals because his goal of playing with his children does not require them. I can adjust exercise intensity; investigating worsening knee pain requires referral. This approach follows guidance to tailor activity to a person’s knee symptoms and goals. (National Institute for Health and Care Excellence [NICE], 2022).

My second decision was between repeating a full-body session three times weekly and varying the emphasis across lower-body, upper-body and functional sessions. I chose varied emphases to make sessions easier to follow and distribute lower-body loading across the week, applying gradual overload while allowing recovery between repeated knee demands. High-box sit-to-stands and shallow leg presses practice controlled knee bending; glute bridges and light hip hinges build hip strength, with an unloaded option if his back becomes uncomfortable. Supported step taps precede low step-ups because stair descent causes pain. Rows and presses cover upper-body movements. I selected these exercises to build strength and function while adapting to his knee symptoms (Goh et al., 2019); diagnosis and clinical management remain with his healthcare team.

I prescribed two sets for most exercises, generally 10–12 repetitions at RPE 4–5, so Mark could learn technique with repetitions in reserve. (American College of Sports Medicine [ACSM], 2009). I increased aerobic duration before adding a third set to the first two lifts in week 4. I considered automatic weekly load increases, but his tolerance is unknown. From week 5, resistance increases by at most 5% only after two tolerated exposures at the top of the rep range with good technique and RPE <=6. Knee pain that increases, pain elevated for more than two hours, or next-day swelling means reducing range or load and repeating the last tolerated dose instead of progressing.

The case reports BMI 32.4, yet 91 kg and 178 cm calculate to 28.7; I would verify the measurements before tracking BMI. I would record symptoms and exercise tolerance, discuss concerns with my supervisor, and refer persistent pain or glucose concerns to his GP or an accredited exercise physiologist. Medication and glucose management remain with his healthcare team.

REFERENCES

National Institute for Health and Care Excellence. (2022, October 19). Recommendations | Osteoarthritis in over 16s: diagnosis and management | Guidance | NICE. https://www.nice.org.uk/guidance/ng226/chapter/Recommendations

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